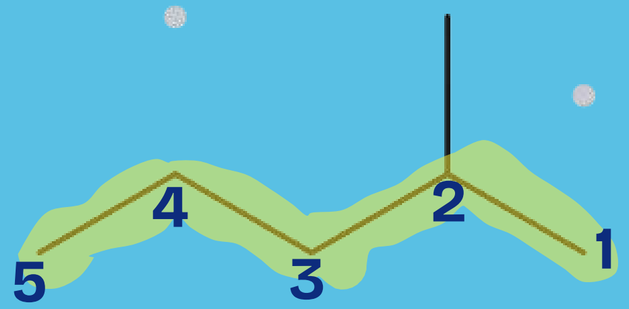
# CARD AO1 - STRUCTURE → NAME

## FOUNDATION LEVEL: MOLECULAR FORMULA (COUNTED): $C_6H_{14}$

### 1. NUMBERED PARENT



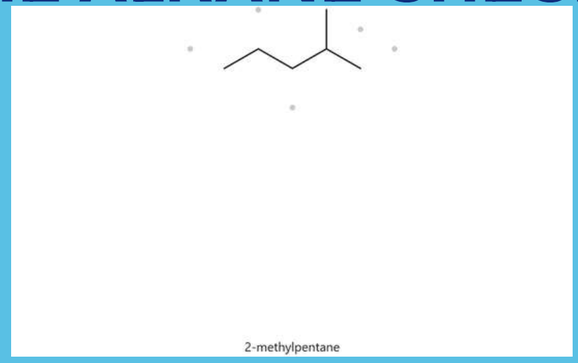

### 2. NAMING TRAIL

- Parent: Longest chain has 5 carbons, so the parent is pentane.
- Numbering: Number from the right end. The branch is at carbon 2 instead of carbon 4.
- Substituents: One methyl group at carbon 2.

### 3. MY NAME

**2-methylpentane**

### 4. THE ALKANE CHECK



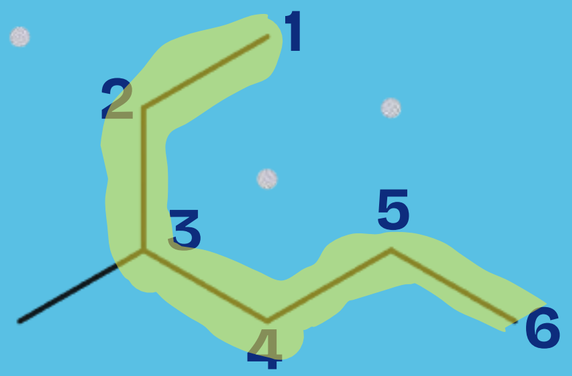

**returned name: 2-methylpentane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD AO2 - STRUCTURE → NAME

## FOUNDATION LEVEL, MOLECULAR FORMULA (COUNTED): $C_7H_{16}$

### 1. NUMBERED PARENT



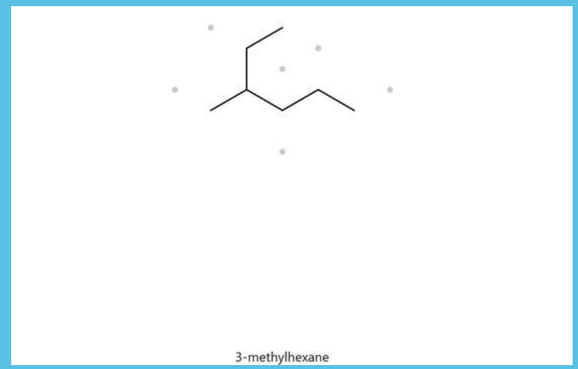

### 2. NAMING TRAIL

- **Parent: The longest continuous chain has 6 carbon atoms, following the bent path through the upper branch. The parent is hexane.**
- **Numbering: Number from the upper-right end. The branch is at carbon 3 instead of carbon 4.**
- **Substituents: One methyl group at carbon 3.**

### 3. MY NAME

**3-methylhexane**

### 4. THE ALKANE CHECK



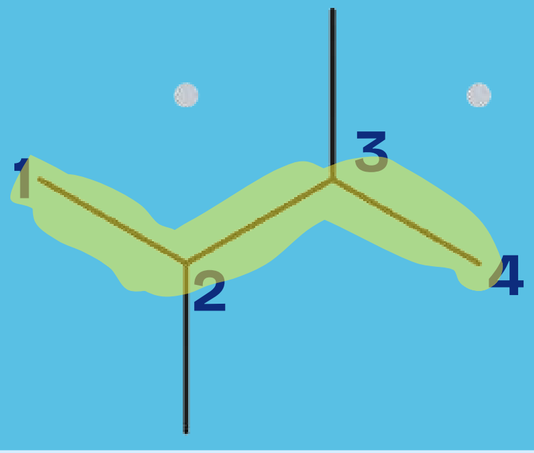

**returned name: 3-methylhexane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD AO3 - STRUCTURE → NAME

## FOUNDATION LEVEL: MOLECULAR FORMULA (COUNTED): $C_6H_{14}$

### 1. NUMBERED PARENT



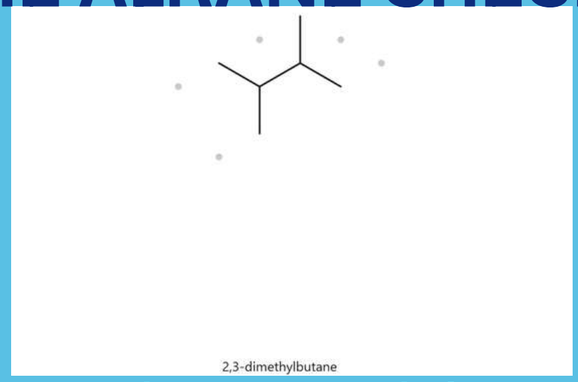

### 2. NAMING TRAIL

- Parent: The longest chain has 4 carbon atoms, so the parent is butane.
- Numbering: Either direction gives the same locants, 2 and 3.
- Substituents: Two methyl groups at carbons 2 and 3. Use the prefix di-.

### 3. MY NAME

**2,3-dimethylbutane**

### 4. THE ALKANE CHECK



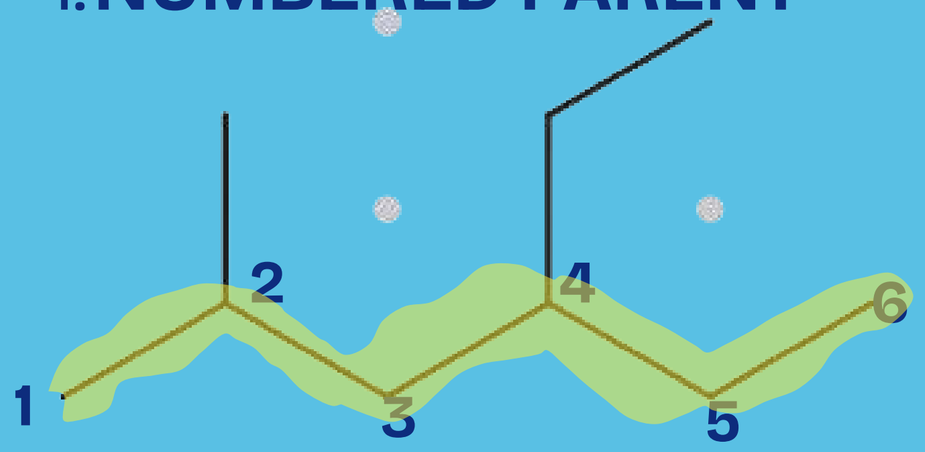

**returned name: 2-methylpentane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A04 - STRUCTURE → NAME

**DEVELOPING LEVEL: MOLECULAR FORMULA (COUNTED): $C_9H_{20}$**

## 1. NUMBERED PARENT



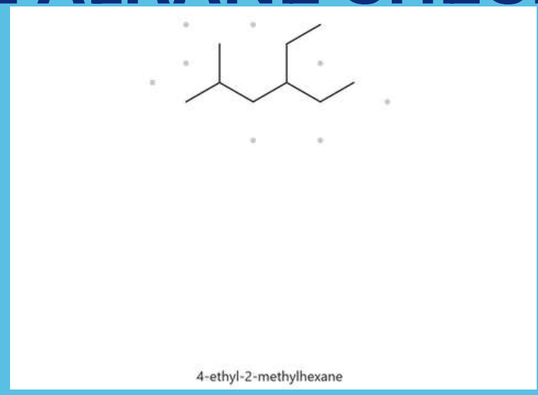

## 2. NAMING TRAIL

- **Parent: The longest chain has 6 carbon atoms, so the parent is hexane.**
- **Numbering: Number from the left end. The locants are 2 and 4, which are lower than 3 and 5 when numbered from the opposite end.**
- **Substituents: One ethyl group at carbon 4 and one methyl group at carbon 2. Write ethyl first because it comes before methyl alphabetically.**

## 3. MY NAME

**4-ethyl-2-methylhexane**

## 4. THE ALKANE CHECK



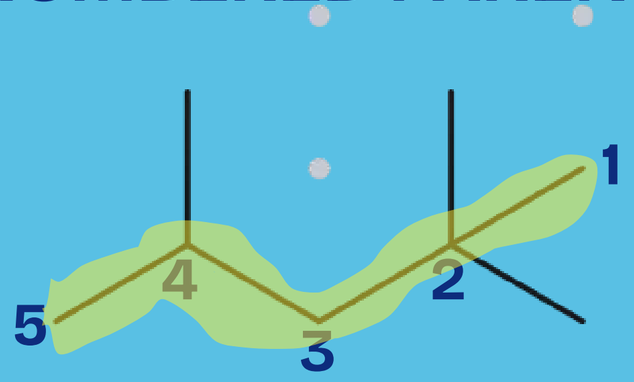

**returned name: 4-ethyl-2-methylhexane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD AO5 - STRUCTURE → NAME

**DEVELOPING LEVEL: MOLECULAR FORMULA (COUNTED): $C_8H_{18}$**

## 1. NUMBERED PARENT



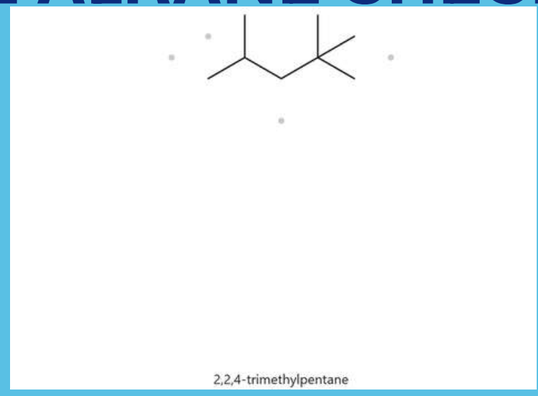

## 2. NAMING TRAIL

- **Parent: The longest chain has 5 carbon atoms, so the parent is pentane.**
- **Numbering: Number from the right end. This gives locants 2, 2, and 4 instead of 2, 4, and 4.**
- **Substituents: Three methyl groups at carbons 2, 2, and 4. Use the prefix tri-.**

## 3. MY NAME

**2,2,4-trimethylpentane**

## 4. THE ALKANE CHECK



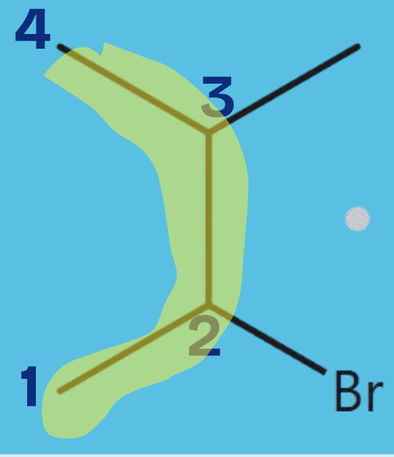

**returned name: 2,2,4-trimethylpentane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A06 - STRUCTURE → NAME

**DEVELOPING LEVEL: MOLECULAR FORMULA (COUNTED): $C_5H_{11}BR$**

## 1. NUMBERED PARENT



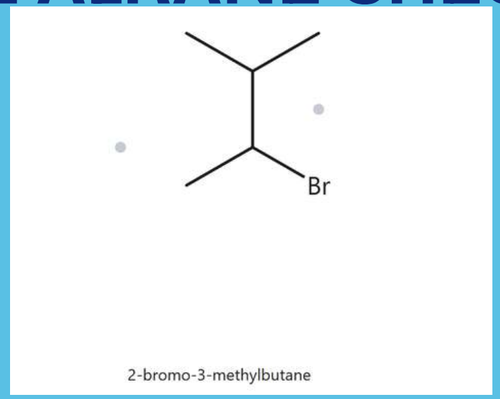

## 2. NAMING TRAIL

- **Parent: The longest chain has 4 carbon atoms, so the parent is butane.**
- **Numbering: Both directions give locants 2 and 3. Break the tie alphabetically: bromo comes before methyl, so bromine receives the lower number.**
- **Substituents: One bromo group at carbon 2 and one methyl group at carbon 3.**

## 3. MY NAME

**2-bromo-3-methylbutane**

## 4. THE ALKANE CHECK



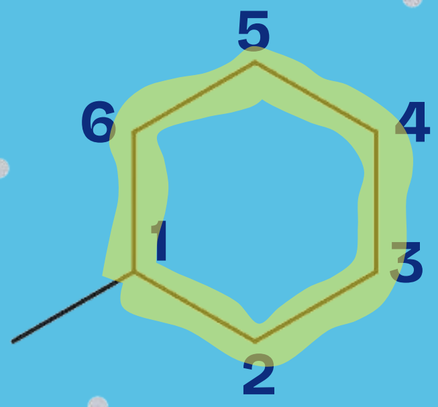

**returned name: 2-bromo-3-methylbutane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A07 - STRUCTURE → NAME

## DEVELOPING LEVEL: MOLECULAR FORMULA (COUNTED): $C_7H_{14}$

### 1. NUMBERED PARENT



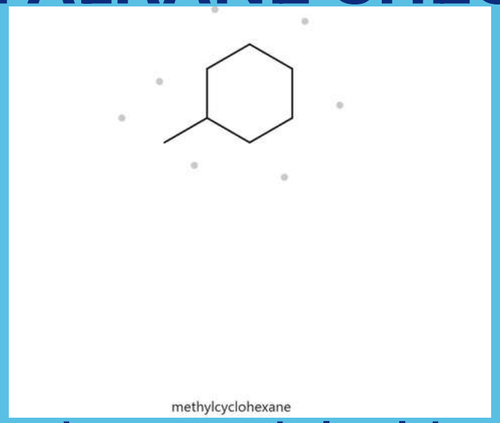

### 2. NAMING TRAIL

- **Parent: The ring has 6 carbon atoms, so the parent is cyclohexane.**
- **Numbering: There is only one substituent, so a locant is unnecessary.**
- **Substituents: One methyl group.**

### 3. MY NAME

**methylcyclohexane**

### 4. THE ALKANE CHECK



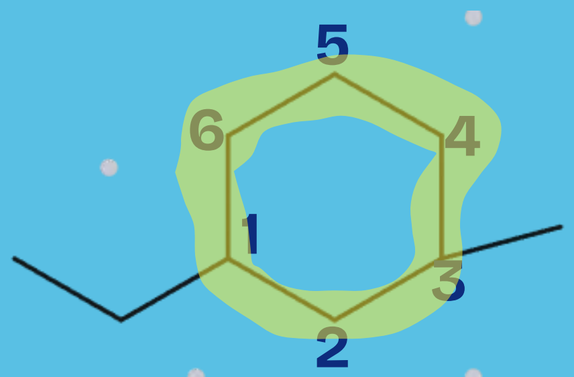

**returned name: methylcyclohexane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A08 - STRUCTURE → NAME

**CHALLENGE LEVEL: MOLECULAR FORMULA (COUNTED): $C_9H_{18}$**

## 1. NUMBERED PARENT



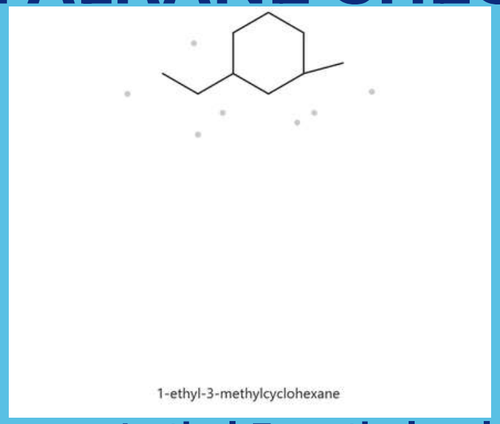

## 2. NAMING TRAIL

- **Parent: The ring has 6 carbon atoms, so the parent is cyclohexane.**
- **Numbering: The substituents have locants 1 and 3 in either direction. Give carbon 1 to ethyl because it comes before methyl alphabetically.**
- **Substituents: One ethyl group at carbon 1 and one methyl group at carbon 3**

## 3. MY NAME

**1-ethyl-3-methylcyclohexane**

## 4. THE ALKANE CHECK



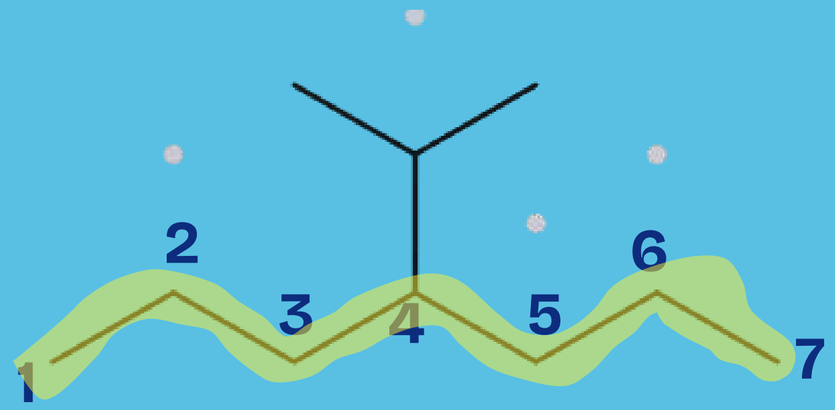

**returned name: 1-ethyl-3-methylcyclohexane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A09 - STRUCTURE → NAME

**CHALLENGE LEVEL: MOLECULAR FORMULA (COUNTED): $C_{10}H_{22}$**

## 1. NUMBERED PARENT



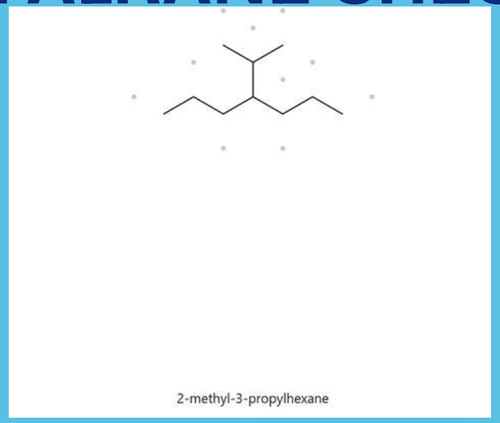

## 2. NAMING TRAIL

- **Parent: The longest continuous chain has 7 carbon atoms, so the parent is heptane.**
- **Numbering: Number from either end. Both directions give the substituent at carbon 4.**
- **Substituents: One propan-2-yl group at carbon 4. The substituent contains 3 carbon atoms and is attached to the parent chain through its second carbon.**

## 3. MY NAME

**4(propan-2-yl)heptane**

## 4. THE ALKANE CHECK



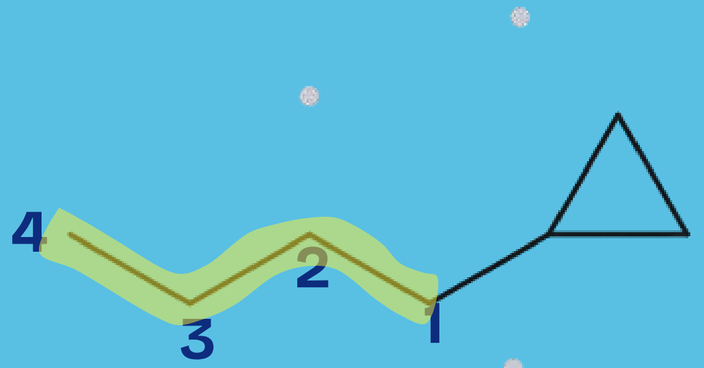

**returned name: 2-methyl-3-propylhexane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A10 - STRUCTURE → NAME

## CHALLENGE LEVEL: MOLECULAR FORMULA (COUNTED): $C_7H_{14}$

### 1. NUMBERED PARENT



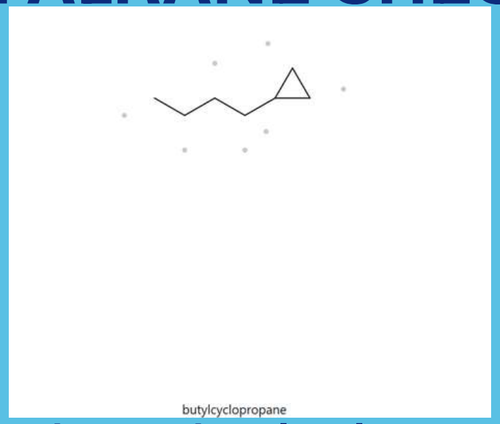

### 2. NAMING TRAIL

- **Parent: The open chain has 4 carbon atoms, while the ring has 3. Therefore, the parent is butane.**
- **Numbering: Number from the end nearest the ring, placing the substituent at carbon 1.**
- **Substituents: One cyclopropyl group at carbon 1.**

### 3. MY NAME

**1-cyclopropylbutane**

### 4. THE ALKANE CHECK




**returned name: butylcyclopropane**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted